ANGLO-CHINESE JUNIOR COLLEGE

JC2 Preliminary Examination

Higher 2

GEOGRAPHY **9173/01**

Paper 1 **26 August 2024**

3 hours

Additional Materials: Insert
Cover Sheet
Answer Paper

Answer Scheme

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Section A

Answer **both** questions.

Cluster 1: Development, Economy and Environment

1 Resource 1 shows the map of Nike production factories. Resource 2 shows the Nike commodity chain. Resource 3 shows the factors affecting Nike's global production network. Resource 4 shows Nike's sustainability score based on the Sustainable Development Goals (SDGs).

(a) Describe the global distribution of Nike factories shown in Resource 1. [4]

Point marked
Award 1 mark for every description to a maximum of 4 marks.

Nike factories can be found in every continent.
Nike factories are predominantly concentrated in Southeast Asia, particularly in countries like Vietnam, Indonesia, and Thailand.
There is also a large number of factories in East Asia with large concentration of workers in China.
However even though there are factories in both Europe and Africa, they are only focused in Western Europe and South Africa.

(b) Using Resource 2, explain how Nike's commodity chain shows the interconnectedness within the global economy. [6]

Point marked
Award 1 mark for every explanation to a maximum of 6 marks.
Award 1 mark for further development of each explanation.

Design & Development: Nike does it's own design and development within it's own headquarters at Oregon and it then looks to outsource marketing by working together with external firms to decide on appropriate marketing for each region.

Global sourcing of materials: Resource 2 shows that Nike outsources it's production and sourcing of raw material to subcontractors from Taiwan and Korea. This reflects the interconnectedness of regions and companies.

Cross-border production: The transformation of these materials into products occurs in multiple countries, particularly in factories across Asia, such as in Vietnam and China. These countries are essential in Nike's production process due to their large manufacturing capacity, demonstrating the company's reliance on global manufacturing hubs.

International distribution: Once products are manufactured, Resource 2 shows that Nike transports these goods to global markets trading companies, which are strategically located to ensure efficient delivery to various consumer markets. This illustrates how Nike connects different parts of the world.

Retail and sales: The commodity chain in Resource 2 reveals that Nike's products are sold in global markets, with retail locations spanning from North America to

Europe and Asia. The fact that products move from factories in Asia to retail outlets worldwide emphasizes the global flow of goods and services.

The resource shows how Nike's global commodity chain integrates various regions through the flows of materials, production, distribution, and consumption, highlighting the deep interconnectivity within the global economy.

(c) Explain the impact of labour unions on transnational companies (TNCs). [4]

Award 1 mark for each relevant point explaining the impact of labor unions on transnational companies (TNCs), with a maximum of 4 marks. Award up to 1 additional mark for further development or elaboration of key points.

Increased labor costs: Labor unions often demand higher wages, better working conditions, and additional benefits for workers. For TNCs, such as Nike, complying with union demands may lead to increased operational costs due to higher wages or improved workplace standards.

Improved working conditions: Labor unions can advocate for better health and safety conditions in factories, pushing TNCs to improve workplace environments. This may require TNCs to invest more in safety protocols, training, and equipment, leading to additional costs but possibly improving worker productivity and satisfaction.

Production disruptions: Labor unions have the power to organize strikes and protests, which can lead to production delays and disruptions. For TNCs relying on global production networks, such disruptions may halt manufacturing and delay product distribution, resulting in financial losses and unmet market demand.

Relocation of production: TNCs may respond to strong union activities by relocating factories to regions with weaker unions or lower labor protection. This can result in job losses in unionized regions but allow TNCs to maintain lower labor costs and more flexible operations.

(d) Using Resource 3, explain the factors that affect Nike’s regional headquarters locations. [6]

Award 1 mark for each explanation of a factor influencing Nike's decision on where to locate it’s RHQ, with a maximum of 6 marks. Award up to 1 additional mark for further development of explanation

Political Stability: Nike chooses to place its RHQs in regions with political stability to minimize risks of operational disruptions. Resource 3 shows that political stability is crucial in ensuring smooth business operations and facilitating consistent communication with local governments, ensuring that Nike's operations face fewer regulatory or legal uncertainties. Political stability helps RHQs to maintain long-term planning and effective management of supply chains.

Economic Factors: Nike locates its RHQs in economically prosperous regions that provide access to high consumer demand and favorable tax policies. This economic advantage allows the company to maximize its profit margins by benefiting from lower operational costs while being closer to important markets, as

indicated in Resource 3. This economic advantage allows Nike to effectively manage regional distribution and sales operations from these strategic locations.

Social and Cultural Considerations: Nike places its RHQs in regions where there is a skilled workforce and an appreciation for global brands. Social factors, such as lifestyle and fashion trends, influence RHQ locations to allow RHQs to monitor and manage real-time data, enabling better decision-making and operational efficiency in product distribution and market analysis.

Technological Infrastructure: RHQs are established in cities with strong technological and communications infrastructure to facilitate smooth management of the supply chain and real-time data analysis. Resource 3 highlights how Nike benefits from advanced logistics and technology to streamline its operations globally.

Legal and Regulatory Environment: Nike chooses locations with favorable legal and regulatory frameworks that support international businesses. These locations ensure that Nike can operate efficiently while complying with local laws and reducing the risk of litigation or regulatory challenges, as highlighted in Resource 3.

Environmental Factors: Nike also considers environmental sustainability when choosing RHQ locations, as the company's commitment to sustainable practices, as seen in Resource 3, where Nike locates RHQs in regions where environmental policies support sustainability efforts, allowing them to integrate eco-friendly practices into regional operation.

(e) Explain the impacts of Nike on the host economy as seen in Resource 4. [6]

Award 1 mark for each explanation of the impacts of Nike on the host economy, with a maximum of 6 marks with the use of data from Resource 4.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Job Creation and Economic Growth: Nike contributes significantly to employment in host countries by creating jobs, particularly in manufacturing sectors. However, according to Resource 4, SDG (Decent Work and Economic Growth) shows a negative score of -2.42, highlighting that despite creating jobs, the quality of employment may be low, with concerns over wages and labor rights. This may limit the long-term economic development potential in these regions.

Tax Revenues for Public Investment: Nike's operations generate tax revenues for the host government, which can be invested in public services such as education and healthcare. Resource 4 illustrates Nike's contribution to SDG (No Poverty), with a positive score of +1.21, as these tax revenues can help alleviate poverty by funding social programs.

Positive Impact on Health and Well-Being: According to Resource 4, SDG (Good Health and Well-Being) has a positive score of +3.33, indicating that Nike's efforts in improving working conditions and promoting health initiatives have a beneficial impact on workers. This suggests that Nike is making strides in ensuring that workplace condition is good and contributing positively to the host economy by fostering a healthier, more productive workforce.

Negative impact of Climate Action: Nike shows a negative score of -3.79 for SDG (Climate Action) in Resource 4. This suggests that Nike's operations may contribute significantly to carbon emissions and environmental degradation in host countries. The negative impact on climate change could result from the energy-intensive manufacturing processes, transportation, and use of non-renewable resources, which worsen environmental conditions in host economies.

(f) Describe how the structure of a less developed country's economy will be affected by shifts in global manufacturing. [4]

Award 1 mark for each relevant description of how shifts in global manufacturing affect the economy of a less developed country, with a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description.

Shifts in global manufacturing lead to a change from an agriculture-based economy to a manufacturing-based economy.
Many LDCs experience a transition where agriculture becomes less dominant as global manufacturing relocates factories to LDCs take advantage of low-cost labor, spurring growth in industrial output.
Over time, as the manufacturing sector grows, it may lead to the development of the service sector.
Increased industrialization can create demand for services such as transport, finance, and retail, causing a rise in tertiary activities that support both the local population and foreign investors.
However, the economic dependency on foreign investors and global markets can deepen. LDCs may rely heavily on foreign manufacturers for income and employment, making their economies vulnerable to global shifts in demand or competition from other low-cost regions.

2(a)	**Explain the temperature pattern shown in Resource 5.** **[4]**

Temperature pattern

- Relatively constant temperature through the year averaging at 2 6.3^{o}C
- High temperature in March at 27°C, lowest in December at 25.5^{o}C
- Low annual range of 1.5^{o}C

Reasons

- High average annual temperature due to the angle of incidence of sun's ray being close to vertical for most parts of the year due its tropical location close to the Equator resulting in intense heating from the concentrated beams of sun rays.
- Small annual temperature range due to being in the tropics with high angle of incidence of sun's rays and cooling effect from the sea due to its coastal location.
- Higher temperature occurs in March which is result of the sun being close by and likely passing over almost directly overhead, and the lack of clouds as March is still the month with relatively lower rainfall. In December, temperature is low due to the lower angle of incidence of sun's rays as the overhead sun has migrated to the southern hemisphere and centred upon the Tropic of Capricorn.
- From December top March there is less clouds as these are the low rainfall months, and it means less insolation is reflected away. More shortwave radiation is reflected away during the wet season from Massy to November, lowering air temperature.
- With less rainfall more of the heat received at the surface is transferred upwards as sensible heat. In the months where rainfall is higher like May to November, it means that there is higher cloud cover and more of the heat at the ground surface is converted into latent heat of vaporization instead of sensible heat and hence lowering air temperature.

Examiner's comments

Following are conceptual errors

- Temperature is due to Am climate.
- This is due to the country being in a tropical Am climate.
- ITCZ is the cause of the temperature pattern

Phrasing / terminology errors and/CE

- Angle of incidence is ***higher*** in the tropics; not ***lesser/lower***

2(b)	**Explain the rainfall pattern shown in Resource 5.** [6]
	Rainfall pattern: • High annual rainfall of 2799 mm • Short dry season from December to March with no more than 120 mm per month and the lowest rainfall month is February with only 60 mm of rainfall. • Highest rainfall occurs October reaching a high of 370 mm. Reasons • Rainfall formed through convection and convergence rainfall, including the processes of clouds formation – heating, rising air parcels, expansion, adiabatic cooling till dewpoint temperature, cloud formation and thus rainfall. • Explain using Hadley cell circulation and the migration of the overhead sun • Wet season occurs in the middle parts of the year starting around May when the *intertropical convergence zone (ITCZ)* passes over Phang Nga Bay area. Intense heating occurs and convectional and convergence rainfall occurs. With the progress of the ITCZ northwards with the northward migration of the overhead sun, more moist air from the ocean in brought onto the land by the southwest monsoon winds, and ITCZ continues to be northwards of Phang Nga Bay location till November. • By December, with the ITCZ having migrated to the southern hemisphere, monsoon winds reverse and pass over the land of IndoChina before arriving at Phang Nga Bay. Such offshore northeast monsoon winds bring little moisture and hence the dry season for this location.

2(c)	**Describe the main features of tower karst in Resource 6A.** **[3]**
	• Tall, isolated hills with steep flanking slope that appears close to 90° angle • Base of isolated hill in the centre middle ground is narrower than the top of the hill • Surface of slope appears uneven with vertical grooves • The isolated hill in the centre middle ground seems to show an overhang above the basal erosional notch. The overhang shows signs of stalactites. • Cave observable in the right middle ground where the larger mass of limestone stands.

2(d)	**With reference to Resources 6A and 6B, compare the weathering processes likely to be operating in each of the landscapes. [6]**
	Each of the processes should be described, citing evidence from the photographs. Resource 6A – carbonation Evidence that it should be carbonation • Tropical climate of Thailand • Tower karst – indicate carbonate rocks • Stalactites on the isolated tower karst – right middle ground, at the base of the overhanging cliff. Resource 6B – insolation weathering and salt weathering; except only one of these two processes Evidence of insolation weathering • Hot tropical desert – which leads to large diurnal range of temperature • Curved layers of rocks on the upper half of the pedestal rock→ indicates spalling arising from insolation or thermal weathering. Evidence of salt weathering • Hot tropical desert – low rainfall, allows for salt crystals to form and also enlarge • Aveoli or small holes on surface of pedestal rock. Weathering processes should be explained in relation to the climatic condition in the respective location, and in the case of the karst in Resource 6A, both the climatic and geologic factors should be used. Do cite evidence from the photograph to support the choice of weathering processes believed to be operating in each landscape.
2(e)	**Describe the ecosystem services provided by the karst landforms in Resource 6A. [6]**
	Karst landscapes, like those found in **Phang Nga Bay** in Thailand, provide various essential ecosystem services. These services can be grouped into four main categories: **provisioning, regulating, supporting, and cultural services**. **1. Provisioning Services** • **Water Supply**: Karst systems store and regulate groundwater. The limestone formations in karst areas often act as natural aquifers, providing water for nearby communities, agriculture, and tourism. • **Raw Materials**: Karst systems contain limestone, which can be mined for construction materials and cement production. However, sustainable use is critical to avoid environmental degradation. **2. Regulating Services** • **Carbon Sequestration**: Karst landscapes, particularly in limestone regions, can sequester carbon through the chemical weathering of rock, which helps mitigate climate change.

	• **Flood Control**: The complex network of caves, sinkholes, and underground rivers in karst systems aids in regulating surface water, reducing the risk of flooding. • **Water Purification**: The porous nature of karst allows for the filtration of water, which can improve water quality as it passes through the system. • **Coastal Protection**: The karst islands and cliffs in Phang Nga Bay act as natural barriers, protecting coastlines from storms, waves, and erosion. **3. Supporting Services** • **Biodiversity**: Karst landscapes often harbor unique ecosystems with specialized flora and fauna. The limestone caves, islands, and surrounding waters in Phang Nga Bay support a range of species, including bats, birds, marine life, and endemic plants. • **Soil Formation**: Karst areas contribute to soil formation through the gradual breakdown of rock and organic material accumulation, which supports local ecosystems. **4. Cultural Services** • **Tourism and Recreation**: The dramatic limestone cliffs, caves, and islands of Phang Nga Bay attract tourists for activities like kayaking, rock climbing, and sightseeing. The region's natural beauty is a major draw for ecotourism. • **Cultural and Spiritual Significance**: Many karst landscapes have cultural and spiritual importance to local communities. In Phang Nga Bay, some caves and islands have historical and cultural value, with ancient paintings and local folklore linked to them. • **Aesthetic Value**: The scenic beauty of the karst formations in Phang Nga Bay contributes to its aesthetic value, inspiring art, photography, and film (e.g., James Bond Island).
2(f)	**Explain the factors likely to have induced the rockfall shown in Resource 7. [5]** **Climate** • Philippines – likely high rainfall location; likely climate is Am with high rainfall during the wet monsoon season • Likely to be in the path of tropical cyclones which leads to very high input of rainfall • Role of excessive water in slopes – high pore pressure, weight of water, and lubricating effect of water leading to high shear strength and reducing shear strength to result in the rockfall **Relief** • In the left foreground and in the right half of the background of photograph of Resource 7 shows steep slopes. • With steep slopes, gravitational pull into the ground is less compared to the tangential which is very high given the steep slope. • This increases shear stress and when water is added to the slope during the wet season, it is likely to fail. **Tectonics** • Philippines lie in the seismic and volcanic belt of the Pacific

	• Earthquake could also shake the materials on the slope inducing shear stress, and in some parts weaken the rocks, reducing shear strength **Geology** • Exposed cliff face in the left middle ground shows sign of joints and bedding planes. • These are lines of weaknesses where water could enter to lubricate. • The lines of weaknesses would also fail during a seismic event. **Human activities** • Road or highway in the foreground of the photograph • To construct the road, a wedge of the hill slope will have to be removed, and the removed material are often used to increase the extent of the flat area. • The loss of toe support reduces shear strength which the wedge would have provided if the toe support was not removes. • With vehicles running on the highway, vibration from the passing vehicles will also induce shear stress.

Section B

Answer **either** question 3 **or** question 4 and answer **either** question 5 **or** question 6.

Cluster 1 Development, Economy and Environment

3 'Political challenges are the main issue that will limit the progress towards sustainable development.'

Evaluate this statement. [20]

Suggested Approach: Candidates can approach the question by evaluating whether political challenges are indeed the most significant barrier to achieving sustainable development. This can be assessed through a comparison of the influence of political challenges with other factors, such as economic or social challenges. Candidates should analyze how political instability, lack of governance, or conflicting political agendas could hinder the progress towards sustainable development. They could then compare this with other challenges like economic constraints or environmental degradation, using examples from both developed and developing nations to assess how political challenges interact with these other factors. The analysis should also consider the global scale, the role of multilateral agreements, and how varying political contexts across countries contribute to different outcomes in sustainable development efforts.

Indicative Content:

Political Challenges:

Climate Change Politics: One of the key political challenges is the slow and fragmented global political response to climate change. Despite scientific evidence pointing to human-induced global warming, the implementation of effective climate policies has been delayed. Political leaders often prioritize short-term national interests over long-term global goals, complicating efforts to integrate climate considerations into decision-making. This is particularly challenging in international negotiations, where developed and developing countries have conflicting priorities regarding economic growth and environmental responsibility.

International Coordination: Global initiatives like the Kyoto Protocol have struggled to achieve consensus on emissions reductions. One of the major challenges is ensuring that all countries, especially developed ones, contribute fairly to global mitigation efforts. Developing countries often argue that they should be allowed to prioritize economic growth, while developed nations are urged to take greater responsibility due to their historical contributions to greenhouse gas emissions. The uneven distribution of power in global politics adds complexity to these negotiations.

Ethical Choices and Climate Justice: Political decisions about climate change are often entangled with ethical dilemmas. The concept of climate justice emphasizes the disproportionate impact of climate change on poorer nations, which are less responsible for greenhouse gas emissions. Political leaders are increasingly faced with making decisions that not only benefit their own nations but also address global inequalities in vulnerability to climate change.

Economic Challenges:

Economic Growth vs. Environmental Sustainability: Rapid economic growth in countries such as China and India has led to an increase in carbon emissions and higher resource demands, creating tension between economic development and environmental sustainability. Studies highlights that in 2009, global CO2 emissions from fossil fuels reached 30.4 billion tonnes, with emerging economies contributing significantly to this increase.

Energy Demand and Fossil Fuel Dependency: Fossil fuels still account for over 80% of the global energy supply, and their continued use poses a significant barrier to sustainable development. Transitioning to renewable energy sources, while essential, is a costly and complex process, especially for developing countries that lack the necessary infrastructure. The book notes that in 2007, the global per capita energy supply was 1.8 tonnes of oil equivalent, illustrating the heavy reliance on fossil fuels.

Water Scarcity and Economic Development: Water scarcity is an escalating issue, particularly in regions with rapidly growing populations. Scientist forecasts that by 2025, global water withdrawals will increase by 50% in developing countries and 18% in developed ones. This rise in demand threatens agricultural productivity and overall economic stability, particularly in water-stressed regions like North Africa and the Middle East. Studies highlights that 884 million people lack access to safe drinking water, underlining the scale of the issue.

Evaluation:

While political challenges are often the most visible and immediate barriers to sustainable development, they are not necessarily the most significant in all contexts. Economic and social factors can have an equally powerful influence, particularly in resource-poor regions where sustainability initiatives require significant financial investment. However, in many cases, political instability can exacerbate these other challenges, suggesting that while political challenges are not the sole limiting factor, they remain a critical issue. Thus, the statement holds some validity, but it is essential to consider the interplay of various challenges across different contexts and scales.

4 Evaluate the relative influence of the state in shaping the global economy. [20]

Possible Approaches: Candidates could approach the question by making a judgement on whether states have a significant influence on the global economy through a consideration of the relative significance of states vis-à-vis other actors such as labour and multilateral institutions. Candidates could analyse the influence of states on the global economy through their role as regulators of global economic activities and the degree to which states can exert an influence over the operations of firms such as TNCs. Candidates could also approach the question by making a judgement on whether states have a greater influence on the global economy in some contexts than others through a consideration of two or more case studies. Candidates could analyse the contextual factors such as the nature of the global economy and the motivation of states in trying to influence it, which may have contributed to the differences in the significance of states.

Supporting the fact that State plays an influential role:

Regulation and Control: States play a significant role in shaping the global economy through regulatory frameworks, trade agreements, tariffs, and taxation policies. For

example, the US-China trade war illustrates how state decisions directly influence global trade flows and the economy.
Infrastructure and Development: States control infrastructure projects, which are vital to economic growth. For instance, China's Belt and Road Initiative (BRI) demonstrates how state-led infrastructure development can reshape global trade routes and increase economic influence.
State-driven Economies: Some countries, like Singapore and China, have adopted state-driven economic models where government intervention is key to economic success. Singapore's government has successfully positioned the country as a global financial hub through strategic planning and investments in education and technology.

Counter Argument:
Influence of TNCs: Transnational corporations, with their global reach and significant economic power, often exert more influence than individual states. For instance, companies like Apple, Amazon, and Google shape global consumer markets and supply chains, sometimes bypassing national regulations.
Role of Multilateral Institutions: Global institutions such as the WTO and IMF set the rules for international trade and finance, limiting the ability of individual states to control their economies. For example, developing countries may be constrained by IMF loan conditions or WTO trade rules that prioritize global market liberalization.
Labour Movements and Civil Society: In some cases, organised labour or grassroots movements can challenge state control and influence economic activities, particularly in industries dependent on large workforces, such as manufacturing and agriculture.

Evaluation:
The relative influence of the state in shaping the global economy is context-dependent. In state-driven economies like China or Singapore, government intervention plays a dominant role. However, in other contexts, particularly in open-market economies or industries dominated by TNCs, the state's influence may be secondary to that of global corporations or multilateral institutions. The balance of power between the state and other actors varies across sectors, regions, and types of economic activity. Therefore, while states remain key players, their influence is often challenged by other powerful economic actors.

Question	Answer	Marks
5	**'Human activities have little impact on the operation of the drainage basin hydrologic systems in the tropics.'** **Evaluate this statement.** Natural Factors to be covered in the answer would include • Climate – annual total rainfall, seasonality of rainfall, role of temperature in inducing evaporation loss; how all these can impact the workings of the many flows and stores; flows affected include interception, infiltration, percolation, overland flow, baseflow, channel flow, river regime, and stores include soil store, water table, groundwater store. • Vegetation – its effect on soil porosity and permeability affecting the flows and stores • Relief – where steep versus gentle surfaces particularly affect rates of infiltration and the generation of overland flow Human activities that can be included • Deforestation • Urbanisation • Dams • Channelisation • Climate change – impact on higher rainfall intensity, or lower than usual rainfall **Conclusion:** Specify the areas where human activities can make a big difference to the operation of the hydro cycle – interception, infiltration, overland flow, river regime, channel stores, subsurface stores - since it can practically all parts of the hydro cycle system → disagree that human activities have little impact. Natural factors dominate only when human activities are not in operation.	20
6	**Evaluate the factors affecting the formation of fluvial landforms in the humid tropics.** Fluvial landforms that should be explained: • Meandering rivers – point bars, slip-off slope, meander scrolls, river cliffs, ox-bow lakes. • Braided rivers – wide and shallow channel with multiple channels, divided by mid-channel bars and eyots • Deltas, esp river-dominated, wave-dominated, tide-dominated deltas • Knickpoint in river long profiles, from either dams, or from geology factor Meandering and braided features – to be explained through the relevant fluvial processes which would be controlled directly by the discharge and sediment load (size and volume). **Discharge and sediment load**	20

will be determined by **climate, vegetation**, and at times, human activities.

Braided river (climate, vegetation, tectonic factor → affecting discharge and sediment load)

Example: braided pattern of the Brahmaputra River – a result of the combination of high sediment load due to monsoonal climate, tectonic uplift, weak rocks in some parts of the Himalayas, tectonic activity leading to earthquake that cause like slope failure generating a vast amount of sediment entering the Brahmaputra, and causing braiding downstream of the Assam region.

Monsoonal climate with a distinct wet and dry season leads to large changes in river discharge where very high river discharge occurs during the wet season where bankfull and even over bankfull discharge occurs. During the high discharge season, river channel has high energy leading to erosion and is able to transport large sediment load and sediments of large sediment size.

During the dry monsoon season, river discharge is significantly lower. The maximum discharge of Brahmaputra at Pandu near Guwahati was recorded as 72,779 cumec on 23.08.62 and minimum discharge was recorded as 1757 cumec on 22.02.63. The average annual discharge is about 20,000 cumec and average dry season discharge is 4,420 cumec.

During the dry season, with much lower river discharge, and the coarser particles are first to be deposited. This accounts for the braiding pattern of the tributaries north of the banks of the Brahmaputra. The finer particles in the tributaries south of the banks of the Brahmaputra leads to a meandering pattern.

Formation of the braided pattern

- Initial deposit of coarse particles leads to obstruction to channel flow forcing diversion of water to the river banks.
- Riverbanks are thus eroded. Eroded materials from the banks which is usually composed of weak unconsolidated materials and unanchored by vegetation ready collapse and adds to the sediment load of the river.
- With excessive sediments in the channel, and coupled with the lower competence and capacity due to the lower discharge during the dry monsoon season, more sedimentation occurs upstream and downstream of the initial deposits, enlarging mid-channel bars and even forming new mid-channel bars downstream of the initial deposit.
- With the formation of several mid-channel bars or eyots, and erosion of the flanking river banks, the braided river becomes wider and shallower.
- When the wet seasons arrives, all the mid-channel bars can be eroded, except for those that have developed to a high elevation and stabilised by the presence of vegetation.

	The Auranga River in northeast India is an example of a braided stream that occurs during the dry season at the lower section of the river bed. In the wet season, the entire channel is used, and a meandering pattern operates. **Meandering stream** conditions and factors • conditions of increasing discharge downstream arising from high rainfall input (eg. the humid tropics of Af, Am climates) • increasing discharge leads to increasing competence and capacity in a downstream direction arising from ○ higher discharge ○ high cross-sectional area and lower wetted perimeter to result in an efficient shape reflected in the high hydraulic radius. Higher hydraulic radius allows for efficient transport of water and sediment downstream with less loss to frictional drag ○ The sediment load is progressively finer downstream and also reduces frictional drag. The load gets finer downstream due to the process of attrition. Also due to sorting where larger and heavier sediment particles are deposited earlier upstream, and only finer ones are further transported downstream. ○ The finer sediment and higher hydraulic radius together reduce channel roughness leading to increasing energy in a downstream direction. • This increasing efficiency leads to the river developing a swinging thalweg that differentially erodes the opposing banks. Where the thalweg is, energy is the highest, and water is elevated on the concave bank, the water combs down the concave bank leading to erosion. Erosion on the concave bank forms the steep slope termed as the river cliff on the side of the concave bank. • Deposition occurs on the convex bank because flow velocity is lower there. As the water is moving against gravity and riding up the slope of the convex bank, water velocity is reduced, and river energy falls. This results in deposition of load on the convex bank to form the slip-off slope and point bars. Point bars usually consist of a series of ridges of sand and gravel. • With differential erosion and deposition on opposing banks, meander migration occurs. The earlier point bars are left behind as the meander bend migrates to form meander scrolls. • In the processes of meander migration, two concave banks may erode to merge into one to form a cut-off forming to result in an ox-box lake. • Such meander features and meander migration are typical of the floodplains of the lower Mississippi river, and the tributaries of the Amazon River, Rio Mamoré and also the Ucayali River. Role of geology • Straight section of channel course in the upper Mekong River – geological control by faults and alignment of folded rocks. • Hard bedrock channels – pothole formation	

- eg. many of which can be found of river bed composed of sandstone of the Mekong River in the area locally named, Samphunbok, in northeast Thailand.
- Also at rapids in humid tropical rivers such as those of the Sungai Dong in Malaysia

- Hard layer of rock forming the caprock and leading to formation rapids and waterfalls, and knickpoint on river long profile. Examples include:
  - Marked variations of gradient occur where streams flow over rocks of differing resistance to erosion, as on the River Nile where the Shabulka Gorge and five cataracts below Khartoum mark the points where the river leaves the sandstones and crosses resistant crystalline rocks.
  - Major waterfalls, such as the Victoria Falls on the Zambesi are caused by belts of igneous rock.
  - The humid lands of the Eastern Highlands of Australia are marked by several falls over the edge of basalt flows, such as the Millstream Falls on the edge of the Atherton Tableland and the Ebor Falls on the Dorrigo Plateau.

Brief description of the main features of deltas

Arcuate
- Fan-shaped, having a rounded, convex outer margin; these are found in the areas where regular longshore drift of other currents keep the seaward edge of the delta trimmed and relatively smooth in shape. Eg. Nile delta

Cuspate
- Pointed like a tooth or cusp shaped by regular but opposing, gentle water movement.

Bird's Foot
- Where the river brings down enormous amounts of fine silt, deposition can occur in a still sea area, along the edges of the distributaries for a vast distance offshore.
- Eg. Mississippi delta

Factors affecting deltas
- Climate – influence the discharge and sediment load
- River flow velocity, sediment load and geomorphic processes (fluvial, tidal, and wave processes)
- Coastal topography
- human activities – dams, sand mining
- Use examples, suggestions: Fly River (tide-dominated, Mississippi (river-dominated) and Nile (wave-dominated)